

Rachel Gross – Curriculum Vitae

University of Virginia, Department of Astronomy
530 McCormick Road, Charlottesville, VA 22904, USA
reg4ff@virginia.edu; ORCID iD: 0000-0002-0477-6047

Academic Appointments

ICI Postdoctoral Fellow, Department of Astronomy, University of Virginia 2026–Present

Education

Ph.D. in Chemistry, University of Virginia May 2026

Dissertation: “UV- and Thermally-Driven Chemistry in Star and Planet Formation: An Experimental, Observational, and Theoretical Approach”

B.Sc. in Chemistry, Minor in Mathematics December 2019

University of Houston-Downtown; *summa cum laude*

Awards and Fellowships

Interdisciplinary Doctoral Fellowship, University of Virginia 2020–2025

First Year Merit Award, UVA Department of Chemistry 2020

Welch Foundation Grant, Scholar’s Academy & Nuclear Regulatory Commission 2019

Publications

- **Gross, R.E.**, Terwisscha van Scheltinga, J., Cleeves, L.I., Dukes, C.A., Garrod, R.T., *et al.*, 2026, *accepted*; arXiv:2609.27031 – “Laboratory Evidence that Methanol-Rich Ice Mantles Lower Methyl Formate Binding Energies.”
- Thompson, W.E., Bergner, J.B., Evans, N.J., *et al.*, including **Gross, R.E.**, 2026, ApJ – “CORINOS V: Radiative Transfer Effects in Protostellar Ice Observations.”
- Turner, A.M., Yang, Y.-L., **Gross, R.E.**, Sakai, N., Kaiser, R.I., 2026, ApJ – “A Chemistry-first Centered Icy Chemical Inventory of Protostellar Sources with JWST.”
- **Gross, R.E.**, Yang, Y.-L., Cleeves, L.I., van Dishoeck, E.F., Garrod, R.T., *et al.*, 2026, ApJ – “CORINOS IV: Quantifying Baseline-fitting Uncertainties in SO₂ Ice Measurements with JWST/MIRI.”
- **Gross, R.E.**, Cleeves, L.I., 2025, ApJ, 980, 189 – “Modeling the Impact of Moderate External UV Irradiation on Disk Chemistry.”
- Williams, B.J., Cleeves, L.I., **Gross, R.E.**, Baker, J., 2025, ApJ, *accepted* – “Investigating the Sulfur Mystery in Protoplanetary Disks Through Chemical Modeling.”
- Ballering, N.P., Cleeves, L.I., Boyden, R.D., McCaughrean, M.J., **Gross, R.E.**, Pearson, S.G., 2025, ApJ, 979, 110 – “Water Ice in the Edge-on Orion Silhouette Disk 114–426 from JWST NIRCам Images.”

Awarded Observational Time

PI: ALMA Cycle 11, 2024.1.00765.S, 9.3 hr – “Uncovering a Hidden Complex Nitrile Reservoir in Planet-Forming Disks.”

Co-I: JWST GO 10076, Cycle 5, 13.0 hr, PI N. Ballering – “I see a little silhouette of a disk: Measuring protoplanetary ice and gas in absorption against a background nebula.”

Co-I: ALMA Cycle 11, 2024.1.00494.S, PI R. Boyden – “The Band 1 Orion Nebula Cluster Disk Survey.”

Co-I: JWST GO 6130, Cycle 3, 17.7 hr, PI R. Boyden – “The Most Typical Planet Formation in the Most Typical Environments.”

Conferences and Presentations

- *Contributed*: DISCO: DISks in COntext Conference, NRAO, Charlottesville, VA, Sep 2026 – “Radiative-transfer modeling of externally irradiated disk ices.”
- *Poster*: Origins of Solar System GRC, South Hadley, MA, June 2025 – “Blended Lines, Clear Insights: Quantifying Weak Ice Features in Protostars with JWST.”
- *Invited*: Center for Astrophysics Institute for Theory and Computation Lunch Series, Cambridge, MA, April 2025 – “JWST MIRI Reveals SO₂ and CH₄ in Planetary Nurseries.”
- *Contributed*: 245th American Astronomical Society, National Harbor, MD, Jan 2025 – “Mid-Infrared Ice Spectroscopy of Class 0 Protostars: Decoding SO₂ and CH₄ Contributions.”
- *Invited*: Exoplanet Coffee Colloquia, University of Wisconsin–Madison, Oct 2024 – “Ice Desorption Experiments with Complex Organics at UVA.”
- *Poster*: New Heights in Planet Formation, ESO, Germany, July 2024 – “Photochemistry and Planet Formation: A Theoretical Study of Irradiated Protoplanetary Disks.”
- *Seminar*: 4th Year Seminar, UVA Chemistry, Mar 2024 – “Photochemistry during Planet Formation: Impacts on the Gas and Ice.”
- *Poster*: Laboratory Astrophysics Workshop (ICE), Kauai, Feb 2024 – “Investigation of COM Desorption from Surfaces and Mixtures of H₂O, CO, and CH₃OH.”
- *Poster*: Origins of Solar System GRC, South Hadley, MA, June 2023 – “Living on the Edge: Disk Chemistry in the Outskirts of Dense Star Forming Regions.”
- *Poster*: Protostars and Planets VII, Kyoto, Japan, Apr 2023 – “Living on the Edge: Disk Chemistry in the Outskirts of Dense Star Forming Regions.”

Teaching, Mentoring, and Outreach

Undergraduate Research Mentor, University of Virginia (2023–Present)

Mentored four undergraduate researchers in laboratory astrochemistry, UHV instrumentation and CAD design, and theoretical chemical modeling.

Teaching Assistant, General Chemistry Lab, UVA (2020–2021)

Instruction and grading for undergraduate general chemistry laboratory courses.

Journal Club Co-Leader, Department of Astronomy, UVA (2023–2024)

Organized and moderated weekly research discussions on current literature in astrochemistry, planet formation, and observational astronomy.

Dark Skies, Bright Kids, Charlottesville, VA (2021–Present)

Enhanced science education through hands-on astronomy activities for elementary and middle school students.

Girls Exploring the Universe, Charlottesville, VA (2022, 2024, 2025, 2026)

Summer astronomy camp for middle school girls, led by UVA women astronomers.